Java vehicle rental system

Bookingcontrollerjava.file ,

```
package com.example.vehiclerental.controller;

import com.example.vehiclerental.entity.Booking;
import com.example.vehiclerental.service.BookingService;
import com.example.vehiclerental.service.CustomerService;
import com.example.vehiclerental.service.VehicleService;
import jakarta.validation.Valid;
import org.springframework.stereotype.Controller;
import org.springframework.ui.Model;
import org.springframework.validation.BindingResult;
import org.springframework.web.bind.annotation.*;

@Controller
@RequestMapping("/bookings")
public class BookingController {
    private final BookingService bookingService;
    private final VehicleService vehicleService;
    private final CustomerService customerService;

    public BookingController(BookingService bookingService, VehicleService
vehicleService, CustomerService customerService) {
        this.bookingService = bookingService;
        this.vehicleService = vehicleService;
        this.customerService = customerService;
    }

    @GetMapping
    public String list(Model model) {
        model.addAttribute("bookings", bookingService.findAll());
        return "bookings/list";
```

```
    }

    @GetMapping("/new")
    public String createForm(Model model) {
        model.addAttribute("booking", new Booking());
        model.addAttribute("vehicles", vehicleService.findAll());
        model.addAttribute("customers", customerService.findAll());
        return "bookings/form";
    }

    @PostMapping
    public String save(@Valid @ModelAttribute("booking") Booking booking,
BindingResult result, Model model) {
        if (result.hasErrors()) {
            model.addAttribute("vehicles", vehicleService.findAll());
            model.addAttribute("customers", customerService.findAll());
            return "bookings/form";
        }
        bookingService.save(booking);
        return "redirect:/bookings";
    }

    @GetMapping("/delete/{id}")
    public String delete(@PathVariable Long id) {
        bookingService.deleteById(id);
        return "redirect:/bookings";
    }
}
```

Customerjavafile,java

```java
package com.example.vehiclerental.controller;

import com.example.vehiclerental.entity.Customer;
import com.example.vehiclerental.service.CustomerService;
import jakarta.validation.Valid;
import org.springframework.stereotype.Controller;
import org.springframework.ui.Model;
import org.springframework.validation.BindingResult;
import org.springframework.web.bind.annotation.*;

@Controller
@RequestMapping("/customers")
public class CustomerController {
    private final CustomerService customerService;

    public CustomerController(CustomerService customerService) {
        this.customerService = customerService;
    }

    @GetMapping
    public String list(Model model) {
        model.addAttribute("customers", customerService.findAll());
        return "customers/list";
    }

    @GetMapping("/new")
    public String createForm(Model model) {
        model.addAttribute("customer", new Customer());
        return "customers/form";
    }
```

```
    @PostMapping
    public String save(@Valid @ModelAttribute("customer") Customer
customer, BindingResult result) {
        if (result.hasErrors()) {
            return "customers/form";
        }
        customerService.save(customer);
        return "redirect:/customers";
    }

    @GetMapping("/edit/{id}")
    public String editForm(@PathVariable Long id, Model model) {
        model.addAttribute("customer", customerService.findById(id));
        return "customers/form";
    }

    @GetMapping("/delete/{id}")
    public String delete(@PathVariable Long id) {
        customerService.deleteById(id);
        return "redirect:/customers";
    }
}
```

Homecontroller.java

```java
package com.example.vehiclerental.controller;

import org.springframework.stereotype.Controller;
import org.springframework.web.bind.annotation.GetMapping;

@Controller
public class HomeController {
    @GetMapping("/")
    public String home() {
        return "redirect:/vehicles";
    }
}
```

Vehicle controller java ,

```java
package com.example.vehiclerental.controller;

import com.example.vehiclerental.entity.Vehicle;
import com.example.vehiclerental.entity.VehicleStatus;
import com.example.vehiclerental.service.VehicleService;
import jakarta.validation.Valid;
import org.springframework.stereotype.Controller;
import org.springframework.ui.Model;
import org.springframework.validation.BindingResult;
```

```java
import org.springframework.web.bind.annotation.*;

@Controller
@RequestMapping("/vehicles")
public class VehicleController {
    private final VehicleService vehicleService;

    public VehicleController(VehicleService vehicleService) {
        this.vehicleService = vehicleService;
    }

    @GetMapping
    public String list(Model model) {
        model.addAttribute("vehicles", vehicleService.findAll());
        return "vehicles/list";
    }

    @GetMapping("/new")
    public String createForm(Model model) {
        Vehicle vehicle = new Vehicle();
        vehicle.setStatus(VehicleStatus.AVAILABLE);
        model.addAttribute("vehicle", vehicle);
        model.addAttribute("statuses", VehicleStatus.values());
        return "vehicles/form";
    }

    @PostMapping
    public String save(@Valid @ModelAttribute("vehicle") Vehicle vehicle,
BindingResult result, Model model) {
        if (result.hasErrors()) {
            model.addAttribute("statuses", VehicleStatus.values());
            return "vehicles/form";
        }
        vehicleService.save(vehicle);
        return "redirect:/vehicles";
    }

    @GetMapping("/edit/{id}")
    public String editForm(@PathVariable Long id, Model model) {
        model.addAttribute("vehicle", vehicleService.findById(id));
```

```
        model.addAttribute("statuses", VehicleStatus.values());
        return "vehicles/form";
    }

    @GetMapping("/delete/{id}")
    public String delete(@PathVariable Long id) {
        vehicleService.deleteById(id);
        return "redirect:/vehicles";
    }
}
```

Entity system files ,

Booking java file

```
package com.example.vehiclerental.entity;

import jakarta.persistence.*;
import jakarta.validation.constraints.*;
import java.time.LocalDate;

@Entity
@Table(name = "bookings")
public class Booking {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;
```

```java
    @ManyToOne(optional = false)
    private Vehicle vehicle;

    @ManyToOne(optional = false)
    private Customer customer;

    @NotNull
    private LocalDate startDate;

    @NotNull
    private LocalDate endDate;

    @NotNull
    @PositiveOrZero
    private Double totalAmount;

    @Enumerated(EnumType.STRING)
    private BookingStatus status = BookingStatus.PENDING;

    public Long getId() { return id; }
    public void setId(Long id) { this.id = id; }
    public Vehicle getVehicle() { return vehicle; }
    public void setVehicle(Vehicle vehicle) { this.vehicle = vehicle; }
    public Customer getCustomer() { return customer; }
    public void setCustomer(Customer customer) { this.customer = customer;
}
    public LocalDate getStartDate() { return startDate; }
    public void setStartDate(LocalDate startDate) { this.startDate =
startDate; }
    public LocalDate getEndDate() { return endDate; }
    public void setEndDate(LocalDate endDate) { this.endDate = endDate; }
    public Double getTotalAmount() { return totalAmount; }
    public void setTotalAmount(Double totalAmount) { this.totalAmount =
totalAmount; }
    public BookingStatus getStatus() { return status; }
    public void setStatus(BookingStatus status) { this.status = status; }
}
```

Bookingstatus java file

```
package com.example.vehiclerental.entity;

public enum BookingStatus {
    PENDING,
    CONFIRMED,
    CANCELLED,
    COMPLETED
}
```

Customer java ,

```
package com.example.vehiclerental.entity;

import jakarta.persistence.*;
import jakarta.validation.constraints.*;

@Entity
@Table(name = "customers")
public class Customer {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    @NotBlank
    private String fullName;
```

```java
    @Email
    @NotBlank
    @Column(unique = true)
    private String email;

    @NotBlank
    private String phone;

    @NotBlank
    @Column(unique = true)
    private String licenseNumber;

    public Long getId() { return id; }
    public void setId(Long id) { this.id = id; }
    public String getFullName() { return fullName; }
    public void setFullName(String fullName) { this.fullName = fullName; }
    public String getEmail() { return email; }
    public void setEmail(String email) { this.email = email; }
    public String getPhone() { return phone; }
    public void setPhone(String phone) { this.phone = phone; }
    public String getLicenseNumber() { return licenseNumber; }
    public void setLicenseNumber(String licenseNumber) {
this.licenseNumber = licenseNumber; }
}
```

Vehicle java

```java
package com.example.vehiclerental.entity;

import jakarta.persistence.*;
import jakarta.validation.constraints.*;

@Entity
@Table(name = "vehicles")
public class Vehicle {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    @NotBlank
    private String brand;

    @NotBlank
    private String model;

    @NotBlank
    @Column(unique = true)
    private String registrationNumber;

    @NotBlank
    private String type;

    @NotNull
    @Positive
    private Double pricePerDay;

    @Enumerated(EnumType.STRING)
    private VehicleStatus status = VehicleStatus.AVAILABLE;

    public Long getId() { return id; }
    public void setId(Long id) { this.id = id; }
    public String getBrand() { return brand; }
    public void setBrand(String brand) { this.brand = brand; }
    public String getModel() { return model; }
    public void setModel(String model) { this.model = model; }
```

```
    public String getRegistrationNumber() { return registrationNumber; }
    public void setRegistrationNumber(String registrationNumber) {
this.registrationNumber = registrationNumber; }
    public String getType() { return type; }
    public void setType(String type) { this.type = type; }
    public Double getPricePerDay() { return pricePerDay; }
    public void setPricePerDay(Double pricePerDay) { this.pricePerDay =
pricePerDay; }
    public VehicleStatus getStatus() { return status; }
    public void setStatus(VehicleStatus status) { this.status = status; }
}
```

Vehicles java file ,

```
package com.example.vehiclerental.entity;

public enum VehicleStatus {
    AVAILABLE,
    BOOKED,
    MAINTENANCE,
    INACTIVE
}
```

Repository

Booking repository java ,

```
package com.example.vehiclerental.repository;

import com.example.vehiclerental.entity.Booking;
import org.springframework.data.jpa.repository.JpaRepository;

public interface BookingRepository extends JpaRepository<Booking, Long> {
}
```

Vehicle repository

```
package com.example.vehiclerental.repository;

import com.example.vehiclerental.entity.Customer;
import org.springframework.data.jpa.repository.JpaRepository;

public interface CustomerRepository extends JpaRepository<Customer, Long>
{
}
```

Customer java file

```
package com.example.vehiclerental.repository;

import com.example.vehiclerental.entity.Customer;
```

```java
import org.springframework.data.jpa.repository.JpaRepository;

public interface CustomerRepository extends JpaRepository<Customer, Long>
{
}
```

Service

Bookingservice java file

```java
package com.example.vehiclerental.service;

import com.example.vehiclerental.entity.*;
import com.example.vehiclerental.repository.BookingRepository;
import org.springframework.stereotype.Service;
import java.time.temporal.ChronoUnit;
import java.util.List;

@Service
public class BookingService {
    private final BookingRepository bookingRepository;

    public BookingService(BookingRepository bookingRepository) {
        this.bookingRepository = bookingRepository;
    }

    public List<Booking> findAll() { return bookingRepository.findAll(); }

    public Booking save(Booking booking) {
```

```
        long days = Math.max(1,
ChronoUnit.DAYS.between(booking.getStartDate(), booking.getEndDate()));
        booking.setTotalAmount(days *
booking.getVehicle().getPricePerDay());
        booking.setStatus(BookingStatus.CONFIRMED);
        booking.getVehicle().setStatus(VehicleStatus.BOOKED);
        return bookingRepository.save(booking);
    }

    public void deleteById(Long id) { bookingRepository.deleteById(id); }
}
```

customer service file java

```
package com.example.vehiclerental.service;

import com.example.vehiclerental.entity.Customer;
import com.example.vehiclerental.repository.CustomerRepository;
import org.springframework.stereotype.Service;
import java.util.List;

@Service
public class CustomerService {
    private final CustomerRepository customerRepository;

    public CustomerService(CustomerRepository customerRepository) {
        this.customerRepository = customerRepository;
    }

    public List<Customer> findAll() { return customerRepository.findAll();
}
    public Customer save(Customer customer) { return
customerRepository.save(customer); }
```

```
    public Customer findById(Long id) { return
customerRepository.findById(id).orElseThrow(); }
    public void deleteById(Long id) { customerRepository.deleteById(id); }
}
```

Vehicle service file java

```
package com.example.vehiclerental.service;

import com.example.vehiclerental.entity.Vehicle;
import com.example.vehiclerental.repository.VehicleRepository;
import org.springframework.stereotype.Service;
import java.util.List;

@Service
public class VehicleService {
    private final VehicleRepository vehicleRepository;

    public VehicleService(VehicleRepository vehicleRepository) {
        this.vehicleRepository = vehicleRepository;
    }

    public List<Vehicle> findAll() { return vehicleRepository.findAll(); }
    public Vehicle save(Vehicle vehicle) { return
vehicleRepository.save(vehicle); }
    public Vehicle findById(Long id) { return
vehicleRepository.findById(id).orElseThrow(); }
    public void deleteById(Long id) { vehicleRepository.deleteById(id); }
}
```

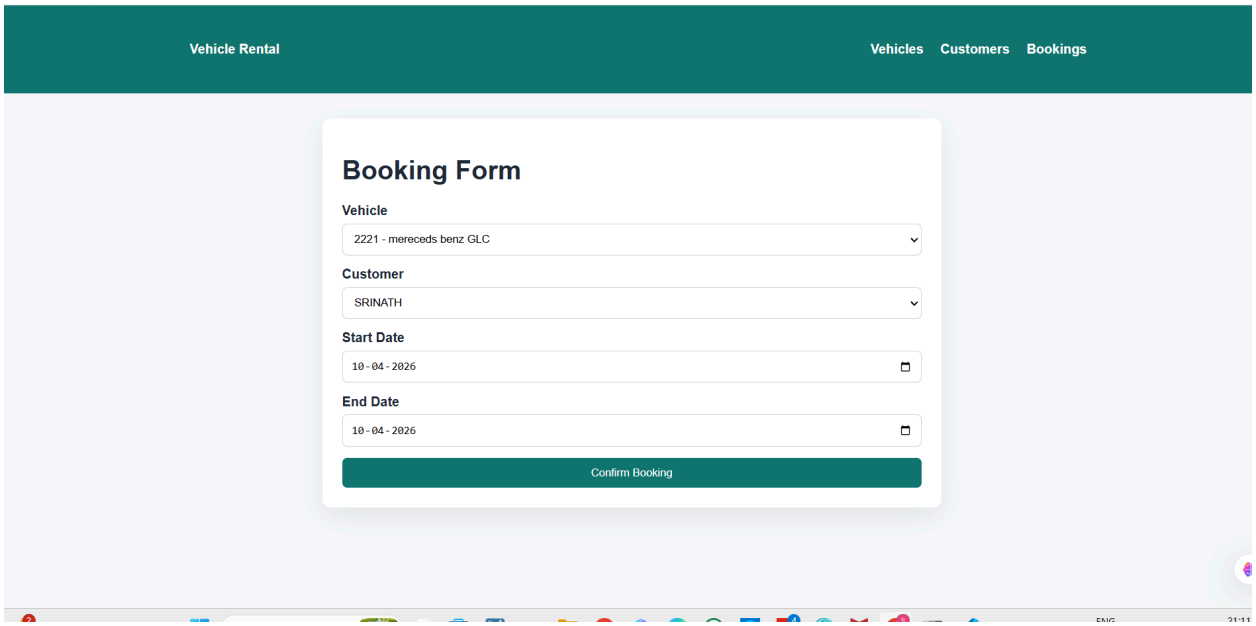
Serviceapplicationjava app ,

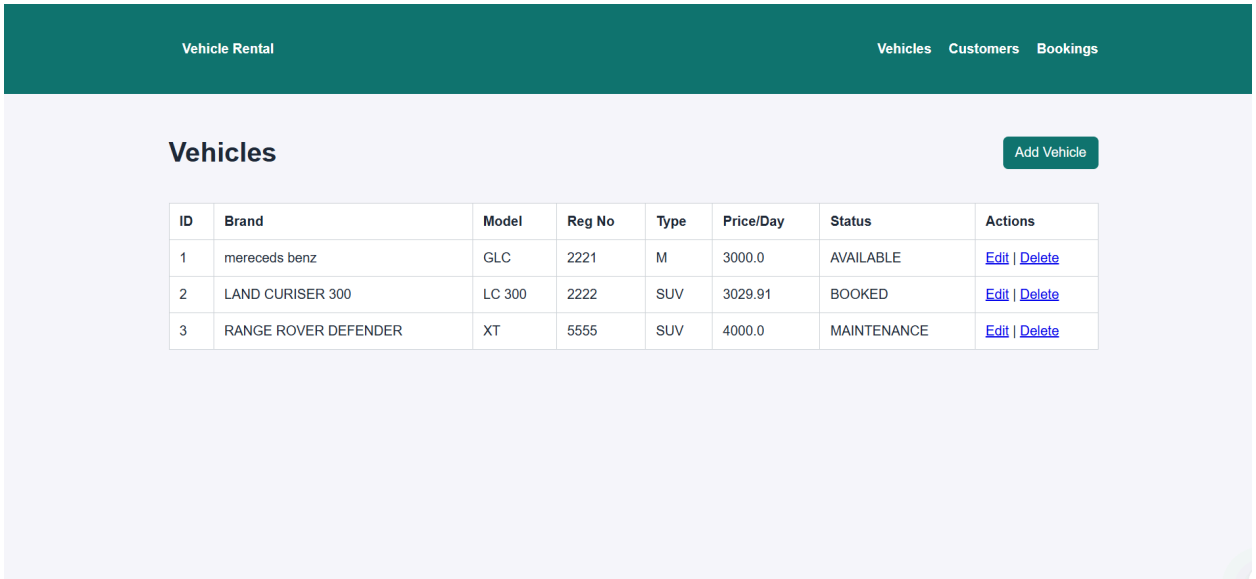
```java
package com.example.vehiclerental;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class VehicleRentalApplication {
    public static void main(String[] args) {
        SpringApplication.run(VehicleRentalApplication.class, args);
    }
}
```

Output s ,





ID	Brand	Model	Reg No	Type	Price/Day	Status	Actions
1	mereceds benz	GLC	2221	M	3000.0	AVAILABLE	Edit | Delete
2	LAND CURISER 300	LC 300	2222	SUV	3029.91	BOOKED	Edit | Delete
3	RANGE ROVER DEFENDER	XT	5555	SUV	4000.0	MAINTENANCE	Edit | Delete

Vehicle Rental

Vehicles Customers Bookings

Bookings

New Booking

ID	Vehicle	Customer	Start	End	Total	Status	Actions

Vehicle Rental

Vehicles Customers Bookings

Booking Form

Vehicle

2221 - mereceds benz GLC

Customer

SRINATH

Start Date

dd-mm-yyyy

End Date

dd-mm-yyyy

Confirm Booking